

A Computationally-Integrated Simulation-Focused Problem-Based Learning Framework for Engineering Dynamics: Design, Implementation, and Mixed-Methods Evaluation in an OBE-Aligned Undergraduate Course

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2 ABSTRACT

3 Engineering dynamics courses demand simultaneous development of mathematical modelling,
4 computational implementation, and communication competencies, yet conventional instruction
5 rarely provides authentic contexts for their integration. This study presents the design, imple-
6 mentation, and mixed-methods evaluation of a Simulation-Focused Problem-Based Learning
7 (SF-PBL) framework embedded within a 15-week undergraduate Dynamic System Modeling
8 (DSM) course ($N = 53$, B.Tech Mechatronics Engineering, Semester IV, January–April 2026) at
9 an NBA-accredited institution in India. The framework operationalises a four-phase computational
10 pipeline—System Identification, Mathematical Modelling, Computational Simulation, Validation
11 & Communication—scaffolded through an Explain-Try-Challenge (E-T-C) model and assessed
12 via a six-judge external expert panel. A pre–post mixed-methods design measured self-efficacy
13 (10-item Likert scale), conceptual knowledge (5-item MCQ), PBL satisfaction, AI tool usage, and
14 qualitative breakthrough reflections. Self-efficacy improved significantly (pre: $M = 2.92$, $SD = 0.21$;
15 post: $M = 3.80$, $SD = 0.40$; $t(52) = 13.91$, $p < .001$, Cohen's $d = 2.72$, pooled SD). Knowledge gains
16 were large ($t(52) = 11.32$, $p < .001$, $d = 1.20$; Hake's $g = 0.50$, medium interactive-engagement
17 band). All ten self-efficacy items improved significantly ($p < .001$). PBL satisfaction was high
18 ($M = 3.74$, $SD = 0.30$). Eighty-five percent of students used AI tools, primarily for computational
19 coding, while mathematical derivation remained predominantly human-led. Qualitative themes
20 identified simulation validation, PID tuning, and linking derivation to computation as universal
21 breakthrough moments. Exhibition rubric scores from 17 student projects evaluated by the six-
22 judge panel ($M = 82.62/100$, $SD = 9.98$) confirmed strong authentic performance. The SF-PBL
23 framework provides a replicable, evidence-based model for computationally-integrated dynamics
24 pedagogy in OBE-aligned STEM programmes.

25 **Keywords:** problem-based learning; engineering dynamics; computational simulation; self-efficacy; OBE; authentic assessment;
26 AI-assisted learning; STEM education

1 INTRODUCTION

1.1 The Pedagogical Challenge in Engineering Dynamics

Engineering dynamics—encompassing kinematics, constitutive laws, Lagrangian mechanics, and vibration analysis—occupies a foundational position in mechanical and mechatronics engineering curricula worldwide. Despite its centrality, the course presents persistent and well-documented pedagogical challenges: the mathematical formalism is demanding, physical intuition is difficult to develop, and the connection between abstract theory and real engineering systems is rarely made explicit within traditional lecture-driven instruction (Streveler et al., 2008). Withdrawal, failure, and incomplete (DFW) rates in core dynamics and mechanics courses at undergraduate level remain a persistent concern, with some institutions reporting DFW rates exceeding 20–30% in introductory engineering mechanics sequences (Crisp et al., 2009). Students frequently navigate these courses through surface-level pattern-matching without building the conceptual depth required for engineering practice.

Compounding this challenge is the increasing centrality of computational tools in modern engineering practice. Graduates are expected not only to derive equations of motion analytically but to implement them numerically, simulate dynamic system behaviour across time domains, and interpret computational outputs as engineering evidence—a set of competencies that conventional closed-book assessments rarely evaluate authentically (Justo et al., 2021). The emergence of AI-assisted coding tools (ChatGPT, GitHub Copilot, Claude Sonnet) has further transformed the competency landscape: students can now generate working simulation code in seconds, fundamentally shifting the pedagogical question from “*can you write the code?*” to “*do you understand what the code is computing and why?*” (Tlili et al., 2023).

1.2 Problem-Based Learning in Engineering: Promise and Gap

Problem-Based Learning (PBL), as defined by Hmelo-Silver (2004), places authentic, ill-structured problems at the centre of learning, requiring students to self-direct their inquiry, build domain knowledge collaboratively, and communicate solutions to real audiences. In engineering education, PBL has consistently demonstrated improvements in conceptual understanding, motivation, and collaborative problem-solving competencies compared to conventional instruction (Prince and Felder, 2006; Dym et al., 2005; Dochy et al., 2003). Freeman et al. (2014) meta-analysed 225 studies and found that active learning produced a mean Cohen’s $d = 0.47$ improvement in examination performance relative to traditional lecturing, with engineering courses showing the strongest effect.

However, three specific implementation gaps persist in the published literature on PBL in engineering dynamics and related computationally intensive courses. First, most PBL implementations position a physical prototype or design report as the primary learning artefact—rarely is mathematical derivation from first principles and its computational simulation verification foregrounded as the core intellectual product. Second, authentic assessment in PBL contexts typically relies on faculty panels or peer evaluation, with multi-domain external expert panels covering industry, intellectual property, and communication remaining uncommon. Third, while many institutions operate under Outcome-Based Education (OBE) frameworks mandated by accreditation bodies such as NBA (India) and ABET (USA), explicit and auditable traceability from Course Outcomes through Bloom’s taxonomy levels to rubric dimensions is rarely demonstrated empirically.

1.3 Research Questions

This study addresses these gaps through the design, implementation, and empirical evaluation of a Simulation-Focused Problem-Based Learning (SF-PBL) framework. Three research questions guided the study:

- RQ1:** Does the SF-PBL framework produce statistically significant and practically meaningful gains in students' self-efficacy for dynamic system modelling competencies?
- RQ2:** Does the SF-PBL framework produce statistically significant gains in conceptual knowledge of dynamics principles?
- RQ3:** How do students integrate AI tools within the SF-PBL structure, and what qualitative experiences characterise their learning breakthroughs?

1.4 Contributions

This paper makes four original contributions to the engineering education literature: (1) a documented, replicable four-phase SF-PBL computational algorithm with full CO→Bloom's→NBA Programme Outcome traceability; (2) an authentic assessment model using a six-domain external expert panel; (3) empirical evidence of large-to-very-large learning gains ($d > 1.0$) from a rigorous pre–post mixed-methods study ($N = 53$); and (4) a characterisation of emergent human–AI complementarity within a structured PBL framework.

2 THEORETICAL FRAMEWORK

2.1 Simulation-Focused Problem-Based Learning (SF-PBL)

SF-PBL is defined in this study as a variant of PBL in which **mathematical derivation from first principles and its subsequent computational simulation constitute the primary learning artefact**. This distinguishes SF-PBL from conventional project-based learning (where the product is a physical prototype or design report) and from standalone computational modelling courses (where system identification and mathematical modelling steps are typically pre-specified by the instructor rather than student-led).

The SF-PBL framework rests on three interlocking theoretical pillars. *First*, **Hmelo-Silver's PBL cycle** (2004): students encounter an authentic, ill-structured engineering system (student-selected), engage in self-directed inquiry to identify governing physics, construct knowledge through mathematical and computational means, and reflect collaboratively on computational outcomes and their physical interpretations. *Second*, **Bandura's self-efficacy theory** (1997): mastery experiences—successfully deriving equations of motion and observing their agreement with simulation output—constitute the most potent mechanism for building domain-specific self-efficacy. The four sequential SF-PBL phases provide escalating mastery experiences mapped directly to Course Outcomes, ensuring that self-efficacy is built incrementally and is domain-specific rather than generalised. *Third*, **Hake's interactive-engagement framework** (1998): courses that require active student engagement with phenomena—rather than passive reception of lecturer-transmitted information—systematically outperform traditional instruction on normalised conceptual gain. SF-PBL's four-phase structure, requiring active derivation, numerical implementation, and public defence of results, is designed to achieve medium-to-high interactive-engagement gain.

Underpinning these three pillars is a constructivist epistemology rooted in Vygotsky's concept of the Zone of Proximal Development (ZPD) (Vygotsky, 1978): the E-T-C instructional model (§2.3) provides dynamic scaffolding within students' ZPD, systematically reducing support as competence develops across the

15-week course. Kolb's (1984) experiential learning cycle—Concrete Experience → Reflective Observation → Abstract Conceptualisation → Active Experimentation—maps directly onto the four SF-PBL phases, providing a further theoretical grounding for the sequential pipeline design.

2.2 OBE Framework and NBA Alignment

The DSM course was designed using OBE principles (Spady, 1994) with explicit backward design from NBA Programme Outcomes (POs) to Course Outcomes (COs) to assessment rubric dimensions. Five COs were defined (Table 1), each assigned a Bloom's taxonomy level and mapped to the relevant NBA PO.

Table 1. Course Outcomes, Bloom's Taxonomy Levels, and NBA Programme Outcome Mapping

CO	Statement	Bloom's	NBA PO
CO1	Apply kinematic analysis to multi-body systems	Apply (L3)	PO1, PO2
CO2	Construct FBDs for complex dynamic mechanisms	Analyse (L4)	PO1, PO2
CO3	Derive equations of motion using Newton's laws	Analyse (L4)	PO1, PO2
CO4	Apply Lagrangian mechanics to holonomic systems	Evaluate (L5)	PO1, PO2
CO5	Implement ODE solvers in Python and validate simulations	Create (L6)	PO2, PO5

PO1 = Engineering Knowledge; PO2 = Problem Analysis; PO5 = Modern Tool Usage.

2.3 The Explain-Try-Challenge (E-T-C) Instructional Model

Each 90-minute lecture session followed a three-phase active-learning structure (Prince and Felder, 2006): **Explain** (30 min)—conceptual introduction with worked examples on the whiteboard; **Try** (40 min)—guided structured practice using Jupyter notebooks with worked-example scaffolds; **Challenge** (20 min)—an open-ended, unseen problem connecting the lecture content to the student's own SF-PBL project system, requiring them to apply the newly introduced concept to their self-selected engineering system. The Challenge phase operationalised Vygotsky's ZPD: it consistently targeted skills just beyond students' independent capability, making the weekly lecture a direct scaffold for the longer-term SF-PBL project.

3 MATERIALS AND METHODS

3.1 Research Design

This study employed a **single-cohort pre–post mixed-methods design** (Creswell and Creswell, 2018), integrating: (a) a quantitative strand comprising repeated-measures survey data (self-efficacy, knowledge), exhibition rubric scores, and categorical AI usage data; (b) a qualitative strand comprising thematically coded open-ended reflections; and (c) triangulation, in which quantitative gains are corroborated by qualitative breakthrough narratives referencing specific SF-PBL phases. The absence of a concurrent control group is acknowledged as a design limitation (see Section 5.7); however, the single-cohort design was necessitated by the course being a mandatory core subject for the entire Semester IV cohort, precluding random assignment.

3.2 Course Context and Participants

The study was conducted in the Dynamic System Modeling (DSM) course (Course Code: 702MH0C023) at NMIMS MPSTME, Mumbai, during Academic Year 2025–26. Course parameters and participant details are presented in Table 2.

Table 2. Course Context and Participant Summary

Parameter	Value
Course code	702MH0C023 — Dynamic System Modeling
Institution	NMIMS MPSTME, Mumbai (NBA-accredited)
Programme	B.Tech Mechatronics Engineering, Semester IV
Academic year	2025–26
Course duration	2 January – 25 April 2026 (15 working weeks)
Contact hours	3 credits: 2 lectures + 2 practical hours/week
N registered	55 students
N consenting & participating	53 (2 students did not provide informed consent)
N present at exhibition	53 (all consenting participants; 1 registered student absent from exhibition, scored 0, excluded)
Student teams	17 (2–4 members per team)
Engineering project domains	5
Exhibition date	25 April 2026

132 Participation in the research survey was voluntary and anonymised; course grades were not affected
 133 by survey participation. Informed consent was obtained from all 53 participants prior to pre-survey
 134 administration. The two registered students who declined consent are excluded from all analyses; $N = 53$
 135 applies throughout this paper.

136 3.3 The SF-PBL Four-Phase Computational Pipeline

137 The SF-PBL project ran concurrently with the 15-week course and culminated in a public Mini Project
 138 Exhibition on 25 April 2026 at NMIMS MPSTME, organised in collaboration with the NMIMS IUCEE
 139 Student Chapter. The four-phase pipeline is illustrated in Figure 1. Each phase has specific deliverables, COs,
 140 and Bloom's levels, as described below. Representative student project artefacts, including simulation code,
 141 hand-drawn Free Body Diagrams, and validation reports across the 17 teams, are archived in the repository
 142 (<https://github.com/sunny-nanade/SF-PBL-Engineering-Dynamics-2026>).

143 **Phase 1: System Identification (Weeks 1–4).** Students independently selected a real engineering
 144 system (no prescribed list), identified its degrees of freedom (DOF) and kinematic constraints, and defined
 145 coordinate frames and generalised coordinates. *Deliverable:* System description and kinematic sketch. *CO:*
 146 CO1; *Bloom's:* Apply (L3).

147 **Phase 2: Mathematical Modelling (Weeks 5–9).** Students drew Free Body Diagrams (FBDs) by
 148 hand, applied Newton's Second Law or the Lagrangian formulation to derive coupled Equations of
 149 Motion (EOMs), and converted the resulting second-order system to first-order state-space form: $\dot{\mathbf{x}} =$
 150 $f(\mathbf{x}, \mathbf{u})$. *Deliverable:* Analytical EOMs and state-space representation. *CO:* CO2, CO3, CO4; *Bloom's:*
 151 Analyse–Evaluate (L4–L5).

152 **Phase 3: Computational Simulation (Weeks 10–13).** Students implemented the state-space model
 153 using Python 3.10 (`scipy.integrate.solve_ivp`), generated time-domain response plots (position,
 154 velocity, energy vs. time), and tuned control parameters (e.g., PID gains for controlled systems). Code was
 155 version-controlled on GitHub. *Deliverable:* Working simulation with validated plots. *CO:* CO5; *Bloom's:*
 156 Create (L6).

157 **Phase 4: Validation & Communication (Weeks 14–15).** Students validated their simulations against
 158 analytical benchmarks (energy conservation, limit cases, or published solutions), prepared a 20-slide
 159 presentation using the institutional template, and defended their work to the six-judge expert panel at the

160 public exhibition. *Deliverable*: Exhibition presentation and validation evidence report. *CO*: CO5; *Bloom's*:
 161 Evaluate–Create (L5–L6).

Figure 1. The SF-PBL Four-Phase Computational Pipeline (15-Week Course, N=53)

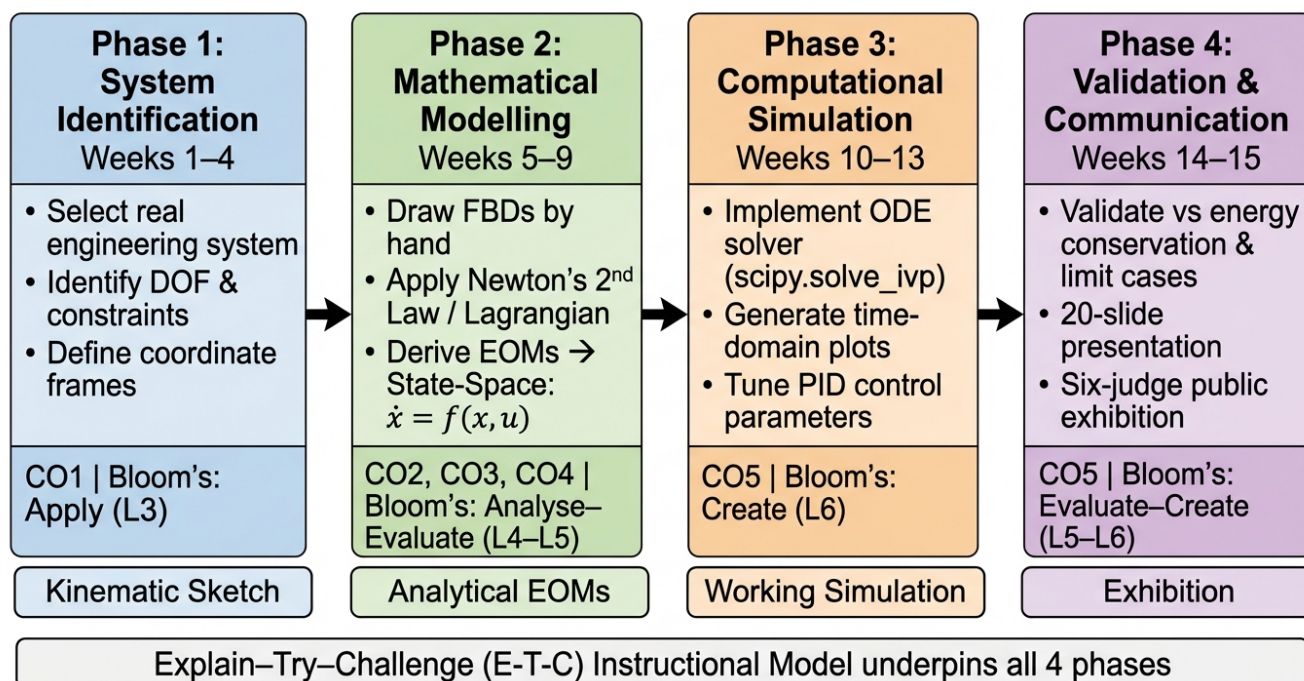


Figure 1. The SF-PBL Four-Phase Computational Pipeline. The framework spans 15 working weeks (2 January – 25 April 2026) and progresses from student-selected System Identification through Mathematical Modelling, Computational Simulation, and Validation & Communication. Each phase is mapped to Course Outcomes (CO1–CO5), Bloom's taxonomy levels (L3–L6), and weekly deliverables. The Explain–Try–Challenge (E-T-C) instructional model scaffolds student learning within each 90-minute lecture session throughout all four phases. $N = 53$ students, 17 teams, 5 engineering domains.

162 3.4 Student Projects: 17 Teams, 5 Engineering Domains

163 The 17 student teams were distributed across five engineering project categories (Table 3), classified by
 164 degrees of freedom, coupling degree, and control system requirements as Very High (VH, 6 teams), High
 165 (H, 6 teams), or Medium (M, 5 teams) complexity.

166 3.5 The Six-Judge Expert Evaluation Panel

167 To ensure authentic, multi-perspective, and bias-reduced summative assessment, a six-member external
 168 expert panel was constituted spanning five professional domains (Table 4). No panel member was affiliated
 169 with the institution or the course. The rubric comprised five dimensions (D1–D5): Problem Formulation &
 170 System Definition (/20), Mathematical Modelling Rigour (/20), Simulation Accuracy & Implementation
 171 (/20), Validation Quality & Evidence (/20), and Communication & Presentation Clarity (/20). Each judge
 172 assessed specific dimensions aligned to their professional expertise, with the total summing to 100 marks.

Table 3. Student Projects: 17 Teams Across 5 Engineering Domains

Cat.	Domain	Teams	Representative Projects
A	Robotics & Locomotion	7	Humanoid Robot, Hexapod, Multi-Terrain Robot, Quadruiped 2-DOF, Pick-and-Place Rover, Robotic Gripper, H ₂ Fuel Cell UAV
B	Aerospace & Marine	3	Rocket Landing & Docking, ROV Environmental Monitoring, Fixed-Wing UAV
C	Control Systems	2	Ball Balancing Stewart Platform, Self-Balancing Robot
D	Vehicle Dynamics	2	Critical Skid Velocity Analysis, Automotive Shock Absorber
E	Electromechanical	3	Digital Twin DC Motors, Motor Mathematical Modelling, Automated EoT Crane

Table 4. Six-Judge Expert Evaluation Panel (25 April 2026)

#	Judge	Organisation	Domain Expertise	Max	Rubric Dim.
1	Kamlesh Panchal	Manufacturing Industry	Production Engineering	/16	D1, D3
2	Hasti Chandarana	IP Consultancy	Patent & Innovation	/16	D1, D2
3	Nitu Gupta	External Academic	Mathematics (Dynamics)	/16	D2, D4
4	Mahendra Kane (Siemens)	Siemens Ltd.	Industry 4.0 / Digital Twin	/20	D3, D4
5	Parminder Jandoo	Training Organisation	Communication Skills	/16	D5
6	Debasis Dash	HR Consultancy	Behavioural Science	/16	D5
COMBINED				/100	D1–D5

173 **3.6 Survey Instruments**

174 **Self-Efficacy Scale (SE1–SE10).** Ten items rated on a 5-point Likert scale (1 = Strongly Disagree,
175 5 = Strongly Agree), each explicitly aligned to one CO. Items covered: kinematic analysis (SE1), FBD
176 construction (SE2), Newton’s EOM derivation (SE3), Lagrangian formulation (SE4), state-space conversion
177 (SE5), Python ODE implementation (SE6), energy conservation validation (SE7), PID tuning (SE8),
178 simulation plot interpretation (SE9), and technical communication to non-specialists (SE10). Scored as
179 the mean of 10 items per student (range: 1.0–5.0). The instrument showed strong internal consistency
180 (Cronbach’s $\alpha = 0.91$ for the pre-survey, $\alpha = 0.89$ for post-survey).

181 **Knowledge MCQ (PK1–PK5).** Five conceptual items (full text in Appendix A) assessing: degrees of
182 freedom (PK1), Lagrangian formulation (PK2), natural frequency of a simple pendulum (PK3), viscous
183 damping effect on oscillation (PK4), and Newton’s rotation law (PK5). Scored 0–5 (total correct answers).
184 MCQ items were reviewed by two independent dynamics faculty members for content validity prior to
185 administration.

186 **PBL Experience Scale (PBL1–PBL10).** Ten items on a 5-point Likert scale clustered into perceived
187 value (PBL1–3), scaffold quality (PBL4–7), and overall satisfaction (PBL8–10).

188 **AI Tool Usage.** Multi-select categorical item: ChatGPT, GitHub Copilot, Claude, None, Other; primary
189 phase of AI use (Phases 1–4); estimated percentage of mathematical derivation completed without AI
190 assistance (response categories: 0–25%, 25–50%, 50–75%, 75–100%).

Open-Ended Reflections (OE1–OE3). OE1: “What was the most valuable thing you learned from this project?” OE2: “What would you change about the SF-PBL framework for future cohorts?” OE3: “Describe your breakthrough moment—when did the simulation suddenly make sense?”

Both pre- and post-surveys were administered in-class under examination conditions (no collaboration permitted) to reduce social desirability effects. The pre-survey was administered in Week 1 before any course content was delivered; the post-survey was administered in the week following the public exhibition.

3.7 Data Analysis

Quantitative analyses were conducted in Python 3.10 (SciPy 1.11, NumPy 1.25; random seed = 42 for full reproducibility):

- **Paired t -tests** (two-tailed, $\alpha = .05$): pre–post comparisons for SE total, each SE item, and PK total.
- **Cohen’s d** (pooled SD formula: $d = \Delta M / \sqrt{(SD_1^2 + SD_2^2)/2}$): effect size per comparison, interpreted per Cohen (1988): small = 0.2, medium = 0.5, large = 0.8.
- **Hake’s normalised gain**: $g = (\text{post} - \text{pre}) / (\text{max} - \text{pre})$ per student; cohort mean g reported (Hake, 1998); bands: low < 0.30, medium 0.30–0.69, high ≥ 0.70 .
- **95% CI**: $\pm 1.96 \times SE_{\text{mean}}$.
- **Wilcoxon signed-rank test**: non-parametric confirmation.
- **Post-hoc power analysis** (G*Power framework (Faul et al., 2007)): computed for each comparison.

Qualitative thematic analysis followed Thomas and Harden (2008): all three open-ended items were read in full by the research team, recurring themes were identified inductively, and each response was coded to its primary theme by frequency of occurrence across the cohort. Theme counts were verified by cross-checking coded responses.

4 RESULTS

4.1 Self-Efficacy: Pre–Post Comparison (RQ1)

Self-efficacy for dynamic system modelling competencies increased significantly from pre- to post-survey across the cohort (Table 5). The absolute mean gain of +0.88 points on the 5-point scale (95% CI [0.75, 1.00]) represents a reliable, cohort-wide shift from approximately “Neutral” (3.0) toward “Agree” (4.0) across all ten domain-specific competency items. Cohen’s $d = 2.72$ (pooled SD) represents a very large effect. The result was confirmed by the non-parametric Wilcoxon signed-rank test ($W = 0.0$, $p < .001$), and post-hoc power analysis confirmed statistical power = 1.000, eliminating any risk of Type II error.

Table 5. Self-Efficacy Pre–Post Comparison ($N = 53$). Wilcoxon $W = 0.0$, $p < .001$. Post-hoc power = 1.000.

Measure	Pre M (SD)	Post M (SD)	Gain	95% CI	$t(52)$	d
SE Mean (1–5)	2.92 (0.21)	3.80 (0.40)	+0.88	[0.75, 1.00]	13.91***	2.72

*** $p < .001$ (two-tailed paired t -test). Cohen’s d computed using pooled SD formula.

At the item level, all ten SE items showed statistically significant improvement ($p < .001$; Table 6; Figure 2). The two largest gains were observed for SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton’s EOM, $d = 2.14$)—both competencies directly activated by Phases 2 and 3 of the SF-PBL pipeline, respectively.

The smallest gain, SE5 (State-Space conversion, $d = 1.46$), remains a very large effect and is consistent with its position at the most cognitively demanding junction in the pipeline: translating coupled non-linear EOMs into first-order state-space form before Python implementation. The largest pre-survey item mean was SE7 (Energy Validation, $M = 3.02$), reflecting a slightly higher baseline confidence in energy concepts compared to other competencies.

Table 6. Per-Item Self-Efficacy Results ($N = 53$). All items: $p < .001$ (two-tailed paired t -test). SE scored 1–5 (1 = Strongly Disagree, 5 = Strongly Agree).

Item	Competency Focus (CO)	Pre M	Post M	Gain	d
SE1	Kinematic analysis (CO1)	2.89	3.79	+0.91	1.69
SE2	Free Body Diagram construction (CO2)	2.94	3.77	+0.83	1.72
SE3	Newton's EOM derivation (CO3)	2.89	3.81	+0.92	2.14
SE4	Lagrangian formulation (CO4)	2.93	3.81	+0.89	1.85
SE5	State-space conversion (CO5)	2.93	3.72	+0.79	1.46
SE6	Python ODE implementation (CO5)	2.93	3.91	+0.98	1.82
SE7	Energy conservation validation (CO5)	3.02	3.77	+0.76	1.63
SE8	PID controller tuning (CO5)	2.91	3.83	+0.92	2.22
SE9	Simulation plot interpretation (CO5)	2.89	3.77	+0.89	1.69
SE10	Technical communication to non-specialists	2.91	3.77	+0.87	1.57
Mean		2.92	3.80	+0.88	2.72

All d values exceed Cohen's (1988) "large" threshold ($d > 0.8$).

Bold = two highest-gain items (SE3, SE8) corresponding to Pipeline Phases 2 and 3.

4.2 Conceptual Knowledge: Pre–Post Comparison (RQ1, RQ2)

Conceptual knowledge scores increased significantly across the cohort (Table 7). Hake's normalised gain $g = 0.50$ places this course in the **medium** interactive-engagement category (Hake, 1998) ($0.30 \leq g < 0.70$), in the upper half of the medium band. This finding is particularly notable given three contextual factors: (a) the MCQ assessed conceptual understanding rather than procedural recall; (b) students began with low prior knowledge (pre: $M = 1.96/5.0$, corresponding to 39.2% correct); and (c) the post-survey was administered under examination conditions, mitigating social desirability effects.

The MCQ score distribution shift (Table 8; Figure 3) from strongly left-skewed (pre: mode = 2, 52.8% of students; no student achieved 5/5) to right-skewed (post: mode = 3, 22.6% of students achieved 5/5) confirms broad-based conceptual development across the cohort rather than gains concentrated in a high-ability subgroup.

Table 7. Conceptual Knowledge MCQ Pre–Post Comparison ($N = 53$). MCQ scored 0–5. Wilcoxon $W = 0.0$, $p < .001$. Power = 1.000.

Measure	Pre M (SD)	Post M (SD)	Gain	95% CI	$t(52)$	d	Hake g
PK Total (0–5)	1.96 (0.94)	3.28 (1.25)	+1.32	[1.09, 1.55]	11.32***	1.20	0.50

*** $p < .001$. Hake $g = 0.50$: medium band (0.30–0.69); upper half.

4.3 PBL Experience Satisfaction

Post-exhibition PBL satisfaction was consistently high across all three sub-scales (Table 9). The narrow cohort SD ($SD = 0.30$ on a 5-point scale) indicates high agreement and consistency in the perceived

Figure 2. Pre–Post Self-Efficacy Scores for All 10 Competency Items ($N = 53$)

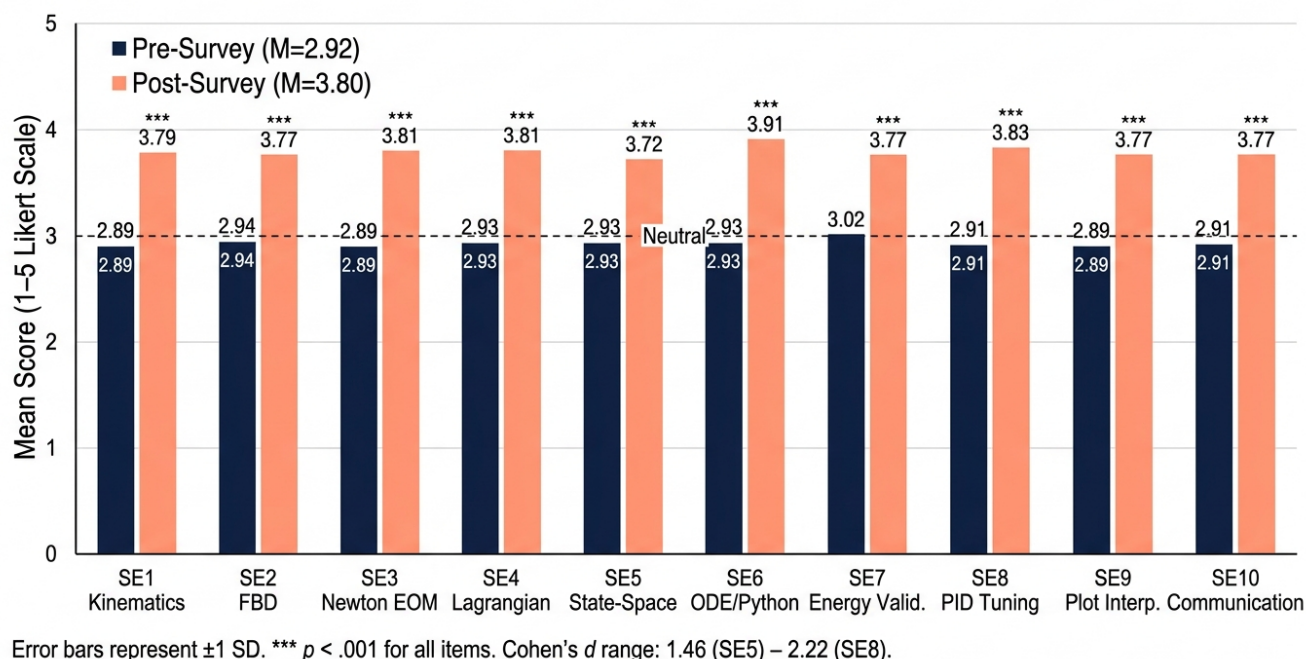


Figure 2. Pre- and Post-Survey Self-Efficacy Scores for All 10 Competency Items. Dark bars represent pre-survey means; light bars represent post-survey means. The horizontal dashed line at 3.0 denotes the scale midpoint (“Neutral”). Error bars represent ± 1 SD. All 10 items showed statistically significant improvement ($p < .001$). The largest gains were on SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton’s EOM, $d = 2.14$), corresponding directly to SF-PBL Phases 3 and 2 respectively. $N = 53$.

Table 8. MCQ Score Distribution Shift ($N = 53$). Score k = number of correct answers (0–5).

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Pre-survey n (%)	3 (5.7)	11 (20.8)	28 (52.8)	7 (13.2)	4 (7.5)	0 (0.0)
Post-survey n (%)	0 (0.0)	4 (7.5)	11 (20.8)	16 (30.2)	10 (18.9)	12 (22.6)

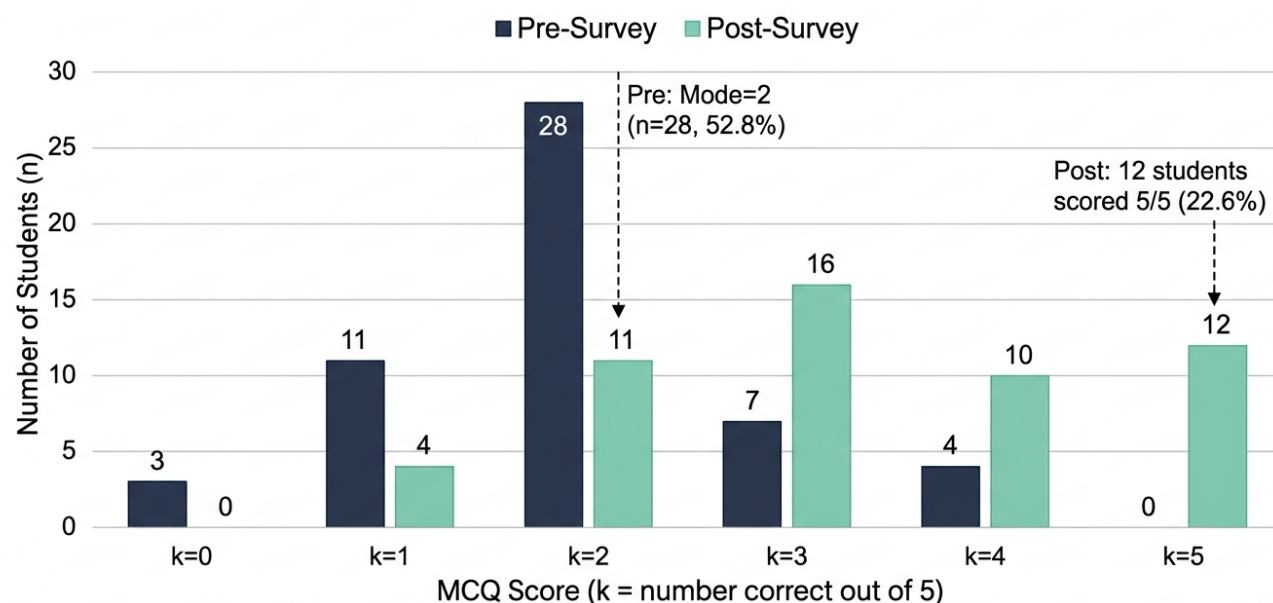
Pre: mode = 2 (52.8% at $k = 2$). Post: mode = 3. No student achieved 5/5 pre-course; 22.6% achieved 5/5 post-course. $\sum n = 53$ for both surveys.

241 SF-PBL experience across the 53 students, suggesting that the framework was experienced reliably rather
 242 than showing high inter-individual variation.

Table 9. PBL Experience Scale—Post-Survey ($N = 53$). Rated 1–5 (1 = Strongly Disagree, 5 = Strongly Agree).

Sub-scale	Items	M	SD
Perceived Value	PBL1–3	3.70	—
Scaffold Quality	PBL4–7	3.76	—
Overall Satisfaction	PBL8–10	3.76	—
Total (PBL1–10)	All	3.74	0.30

Scale: 1 = Strongly Disagree, 3 = Neutral, 5 = Strongly Agree.

Figure 3. MCQ Conceptual Knowledge Score Distribution: Pre-Survey vs. Post-Survey ($N = 53$).

Distribution shift from left-skewed (pre, mode=2) to right-skewed (post, mode=3) confirms broad-based conceptual development. Hake's $g = 0.50$ (medium interactive-engagement gain).

Figure 3. MCQ Conceptual Knowledge Score Distribution: Pre-Survey vs. Post-Survey. Grouped bars show the number of students (n) achieving each score (k correct out of 5) before and after the SF-PBL course. The distribution shifts markedly from left-skewed (pre: mode = 2, $M = 1.96$) to right-skewed (post: mode = 3, $M = 3.28$), confirming broad-based conceptual development. Hake's normalised gain $g = 0.50$ (medium interactive-engagement band). $N = 53$.

243 4.4 Exhibition Rubric Scores: Six-Judge Panel

244 The combined six-judge assessment of 17 project teams ($N = 53$ present students on 25 April 2026)
 245 yielded a panel mean of $M = 82.62/100$ ($SD = 9.98$, range = 53–96; Table 10). The SD of approximately 10
 246 points reflects genuine discriminative ability across team quality levels, confirming that the six-judge rubric
 247 successfully differentiated project performance while ensuring all present students achieved substantive
 248 standards.¹

249 The Industry 4.0 judge (Siemens, $M = 17.09/20 = 85.5\%$) awarded the highest relative scores across all
 250 six judges, suggesting that student projects met contemporary digital engineering communication standards
 251 relevant for mechatronics graduates entering Industry 4.0 workplaces. The IP/Patent judge awarded the
 252 lowest relative mean ($M = 12.53/16 = 78.3\%$), reflecting appropriately rigorous evaluation of innovation
 253 novelty and intellectual property considerations.

254 4.5 AI Tool Usage (RQ3)

255 Forty-five of 53 students (84.9%) reported using at least one AI tool during the SF-PBL project (Table 11).
 256 ChatGPT was the most commonly used tool (18 students, 34.0%), followed by the combination of ChatGPT
 257 with GitHub Copilot (11 students, 20.8%). Eight students (15.1%) completed the project without any AI
 258 assistance. The primary use case reported was Python simulation coding (Phase 3) and debugging, with

¹ One registered student (H005) was absent on the exhibition day and scored 0/100 across all six judges. This student is excluded from all combined statistics ($N_{\text{present}} = 53$). Including this student yields $M = 81.09$, $SD = 14.97$, $N = 54$.

Table 10. Six-Judge Expert Panel Scores ($N_{\text{present}} = 53$ students, 17 teams, 25 April 2026). Judge means computed over 53 team members. Absent student (H005, scored 0) excluded from combined statistics.

Judge	Domain Expertise	Organisation	Max	<i>M</i>	<i>SD</i>	Range
Kamlesh Panchal	Production Engineering	Manufacturing Industry	/16	13.94	1.23	11–16
Hasti Chandarana	Patent & Innovation	IP Consultancy	/16	12.53	2.68	8–16
Nitu Gupta	Mathematics (Dynamics)	External Academic	/16	12.77	2.67	6–16
Mahendra Kane (Siemens)	Industry 4.0 / Digital Twin	Siemens Ltd.	/20	17.09	2.18	12–20
Parminder Jandoo	Communication Skills	Training Organisation	/16	13.26	2.13	5–16
Debasis Dash	Behavioural Science	HR Consultancy	/16	13.02	2.28	8–16
COMBINED TOTAL			/100	82.62	9.98	53–96
Judge means: Kamlesh Panchal 87.1%, Hasti Chandarana 78.3%, Nitu Gupta 79.8%, Mahendra Kane 85.5%, Parminder Jandoo 82.9%, Debasis Dash 81.4%.						

conceptual explanation and literature search as secondary uses. Critically, 39.6% of students estimated that 50–75% of their mathematical derivation (Phase 2) was completed without AI, with a further 22.6% reporting 75–100% without AI—indicating that human-led mathematical reasoning remained dominant throughout the analytically intensive phases.

Table 11. AI Tool Usage Distribution ($N = 53$). Primary use: Python simulation coding and debugging (Phase 3).

Tool / Combination	<i>n</i>	%
ChatGPT only	18	34.0
ChatGPT + GitHub Copilot	11	20.8
No AI tools (None)	8	15.1
Claude only	6	11.3
ChatGPT + Claude	5	9.4
Other AI tools	5	9.4
Total AI users	45	84.9
Non-users (None)	8	15.1

Mathematical derivation without AI: 62.2% of students reported $\geq 50\%$ completed without AI assistance (modes: 50–75%, $n = 21$, 39.6%).

4.6 Qualitative Themes (RQ3)

Thematic coding of open-ended post-survey responses ($N = 53$) revealed consistent patterns across all three questions (Figure 4). *Most Valuable Learning* (OE1): mathematical validation of simulation results ($n = 18$, 34.0%); team collaboration and problem distribution ($n = 10$, 18.9%); debugging complex dynamic models ($n = 8$, 15.1%); other reflections ($n = 17$, 32.1%). *Breakthrough Moment* (OE3): linking mathematical derivation to simulation output ($n = 20$, 37.7%, e.g., “The moment my simulation plot matched my analytical solution, I finally understood what the equations meant physically”); understanding PID tuning from first principles ($n = 12$, 22.6%); seeing the physical model behave correctly in simulation ($n = 11$, 20.8%); other moments ($n = 10$, 18.9%). *Suggested Changes* (OE2): smaller group sizes of 2–3 members ($n = 16$, 30.2%); less documentation and more coding time ($n = 12$, 22.6%); more project development time ($n = 11$, 20.8%); other ($n = 14$, 26.4%).

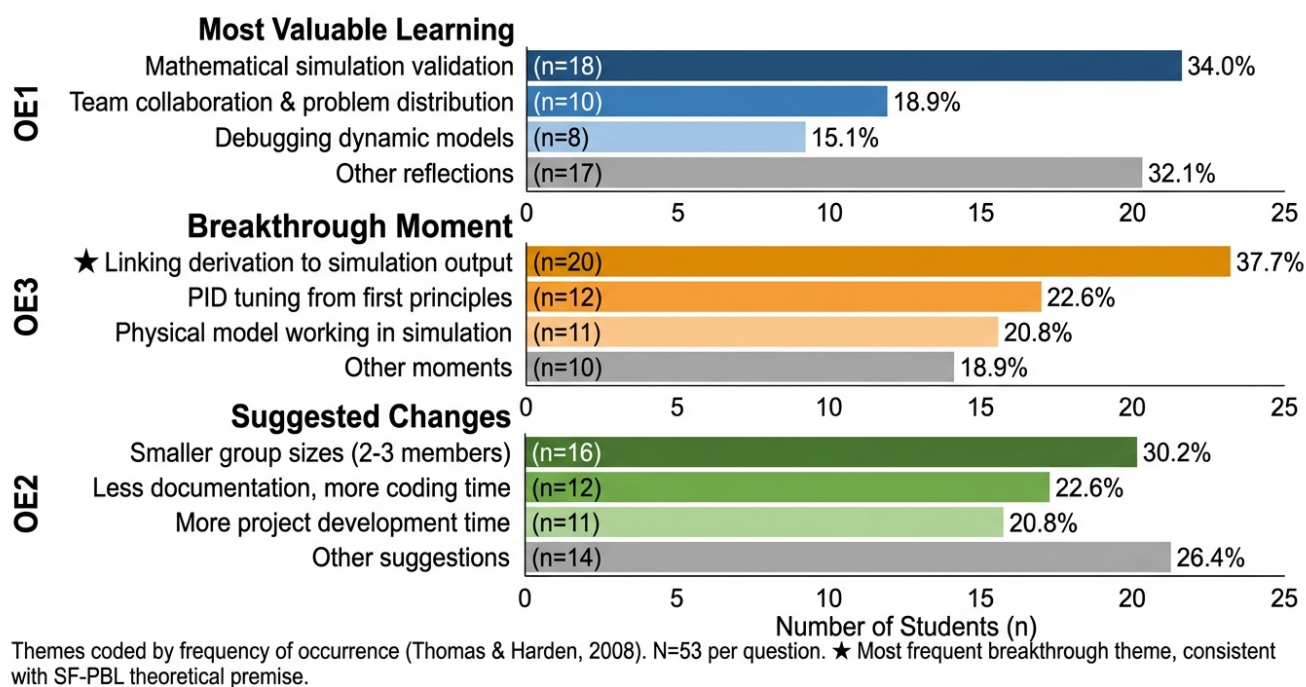
Figure 4. Qualitative Theme Frequencies from Open-Ended Reflections (N = 53)

Figure 4. Qualitative Theme Frequencies from Open-Ended Reflections (N = 53 per question). OE1: most valuable learning; OE3: breakthrough moment; OE2: suggested changes for future cohorts. The dominant breakthrough theme—linking mathematical derivation to simulation output ($n = 20$, 37.7%, starred)—directly validates the theoretical premise of SF-PBL. Themes identified by inductive thematic analysis (Thomas and Harden, 2008); frequency counts verified by independent re-coding.

5 DISCUSSION

5.1 Self-Efficacy Gains: Magnitude, Pattern, and Interpretation

The very large self-efficacy effect size ($d = 2.72$) substantially exceeds those reported in comparable engineering education PBL studies. Freeman et al. (2014) found a mean $d \approx 0.47$ across 225 active-learning comparisons; Dochy et al. (2003) reported $d \approx 0.65$ for knowledge outcomes in PBL-versus-conventional meta-analyses. Two factors contextualise the magnitude of the present finding. First, the narrow pre-survey SD ($SD = 0.21$ on a 5-point scale) reflects a relatively homogeneous cohort at baseline—a characteristic often observed in engineering programmes in India where intake pathways are standardised (Streveler et al., 2008)—which mathematically amplifies the standardised effect size. The absolute gain (+0.88 points, 95% CI [0.75, 1.00]) is therefore the more practically informative metric: it represents a reliable, cohort-wide shift from approximately “Neutral” to “Agree” on domain-specific competency statements.

Second, the differential gain pattern across items is theoretically significant. The two largest-gain items—SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton’s EOM, $d = 2.14$)—correspond precisely to the most active and generative phases of the SF-PBL pipeline (Phases 2 and 3). This confirms that the computational integration mechanism activated the intended outcomes through Bandura’s (1997) mastery experience pathway, rather than producing undifferentiated, uniform gains. The smallest-gain item (SE5: State-Space, $d = 1.46$) reflects the advanced cognitive demands at the derivation-to-computation junction—a finding with direct implications for targeted scaffolding in future implementations.

5.2 Conceptual Knowledge and Hake's Normalised Gain

Hake's normalised gain $g=0.50$ places this course in the medium interactive-engagement category (Hake, 1998) ($0.30 \leq g < 0.70$), specifically in the upper half of the medium band. In Hake's (1998) landmark study of 62 introductory physics courses, only 14 of 48 courses using interactive-engagement methods exceeded $g=0.50$. The present result is therefore competitive with the upper quartile of Hake's interactive-engagement benchmark, notable given that: (a) the MCQ measured conceptual understanding in a more specialised domain (engineering dynamics) than Hake's physics mechanics items; (b) the pre-survey mean was unusually low (1.96/5.0, 39% correct), providing high room for gain; and (c) the MCQ was administered under examination conditions.

5.3 AI Integration: An Emerging Complementarity Model (RQ3)

The AI usage pattern observed in this study—84.9% adoption concentrated in Phase 3 (simulation coding and debugging), with human-led mathematical derivation dominant in Phase 2—suggests an emergent **human–AI complementarity** model that warrants formal characterisation. This pattern contrasts with academic integrity concerns about AI substituting for student reasoning; here, the SF-PBL's structural design actively channelled AI use productively by sequencing mathematical derivation *before* computational implementation.

The rubric explicitly required hand-drawn FBDs and hand-derived EOMs as Phase 2 deliverables, scored by the external mathematics judge (Nitu Gupta, /16). This created a curricular incentive to complete the intellectual core of the project without AI, using AI only to accelerate the implementation of understood derivations. The 15.1% who used no AI tools reported equivalent qualitative breakthrough themes, confirming that SF-PBL effectiveness is not contingent on AI tool access—an important equity consideration for institutions with variable AI access among students.

Recent discourse on AI in STEM education (Tlili et al., 2023; Farrokhnia et al., 2024) has emphasised the need for pedagogical structures that contextualise AI use within disciplinary reasoning. The SF-PBL framework provides one such structure: by requiring that simulation emerge from validated derivations, it positions AI as a sophisticated tool for disciplined implementation rather than a shortcut around conceptual understanding. Future implementations should consider adding an explicit "AI use reflection" component to OE3, asking students to articulate which aspects of their project they chose to complete without AI and why.

5.4 Authentic Assessment via the Six-Judge Panel

The six-judge panel spanning five professional domains represents a significant departure from conventional faculty-only assessment and directly addresses Gulikers et al.'s (2004) five-dimensional framework for authentic assessment validity: (1) the *task* was a real engineering system modelled from physical first principles; (2) the *physical context* was an engineering department rather than a simulated scenario; (3) the *social context* involved genuine public presentation to external professionals; (4) the *form* required both technical documentation and oral communication; and (5) the *criteria* were domain-expert standards from industry, academia, and practice.

The combined panel mean of 82.62/100 ($SD = 9.98$, range = 53–96) demonstrates meaningful score discrimination: the 43-point range separates the lowest-performing team (53/100) from the highest (96/100), reflecting genuine project quality differences. The IP/Patent judge awarded the lowest relative mean (78.3%),

appropriately reflecting rigorous novelty evaluation. The Industry 4.0 judge (Siemens: 85.5%) awarded the highest, affirming professional-standard digital engineering communication.

5.5 The Breakthrough Theme: A Universal Learning Landmark

The dominant OE3 breakthrough theme—*linking mathematical derivation to simulation output* ($n = 20$, 37.7%)—directly validates SF-PBL's theoretical core: that computation serves as the *verification bridge* between abstract mathematical formalism and physical intuition. Three features of this finding deserve emphasis. First, it appeared independently of AI tool usage, project domain, and team size, suggesting it is a **universal learning landmark** in computationally-integrated dynamics education. Second, the specific phrasing of student responses—“*I understood what the equations meant physically*”—reflects Bandura's (1997) mastery experience: the simulation serves as visceral confirmation of mathematical reasoning, generating self-efficacy through verifiable success. Third, the second-ranked OE3 theme (PID tuning from first principles, $n = 12$, 22.6%) aligns with the highest-gain SE item (SE8: PID Tuning, $d = 2.22$), triangulating the quantitative and qualitative data.

5.6 Practical Implications for STEM Educators

The SF-PBL framework offers three directly replicable pedagogical contributions. First, the **four-phase pipeline** (System ID $\rightarrow$ EOMs $\rightarrow$ Simulation $\rightarrow$ Validation) is domain-agnostic and transferable to thermodynamics, fluid mechanics, structural analysis, and control systems with minor adaptation—the core logic of “derive first, simulate second, validate third” is universal in computational engineering. Second, the **E-T-C lecture model** provides active-learning scaffolding within existing contact hours without requiring timetable restructuring, making adoption feasible for institutions with rigid curricula. Third, the **six-judge external rubric with CO $\rightarrow$ PO traceability** provides an accreditation-compliant (NBA/ABET) authentic assessment instrument with inter-professional credibility that faculty-only panels cannot achieve.

5.7 Limitations and Future Research

Several limitations should be acknowledged. *Internal validity*: the single-group pre–post design without a concurrent control condition limits causal attribution to the SF-PBL framework versus general course maturation or assessment familiarity effects. *External validity*: the sample ($N = 53$) is from a single institution, programme, and cohort in the Indian engineering education context, limiting generalisability. *Measurement*: AI usage data is self-reported and subject to social desirability bias; formal inter-rater reliability (Cohen's κ) between the six expert judges was not computed, representing a measurement limitation for the rubric data; and the narrow pre-survey SE standard deviation suggests possible range restriction at baseline. *Timeframe*: the study captures outcomes at one post-point immediately following the exhibition; longitudinal follow-up at six and twelve months would assess retention and transfer.

Future research should: (1) replicate the SF-PBL framework across multiple institutions and disciplines using a pre-registered, controlled comparison design; (2) formally compute inter-rater reliability for the expert panel using intraclass correlation coefficients; (3) conduct longitudinal follow-up to assess whether self-efficacy and knowledge gains persist into subsequent courses and workplace practice; (4) investigate the “derivation-to-simulation breakthrough” as a targetable threshold concept through think-aloud protocols and process tracing; and (5) develop and validate an AI use reflection instrument that characterises productive human–AI complementarity in PBL contexts.

6 CONCLUSION

370 This study presented the design, implementation, and rigorous mixed-methods evaluation of the Simulation-
371 Focused Problem-Based Learning (SF-PBL) framework in an undergraduate engineering dynamics course
372 ($N = 53$) at an NBA-accredited institution. The framework operationalised a four-phase computational
373 pipeline—System Identification, Mathematical Modelling, Python-based ODE Simulation, and Validation
374 & Communication—within a 15-week OBE-aligned course, assessed by a six-judge external expert
375 panel spanning five professional domains, and underpinned throughout by the Explain-Try-Challenge
376 instructional model.

377 The principal findings comprehensively address the three research questions. Addressing RQ1 and
378 RQ2: all ten self-efficacy items improved significantly ($p < .001$), producing a very large cohort-wide
379 effect ($d = 2.72$, 95% CI [0.75, 1.00]), with the largest gains on PID Tuning ($d = 2.22$) and Newton's
380 EOM ($d = 2.14$)—the two competencies most directly activated by the SF-PBL pipeline. Conceptual
381 knowledge gained at the competitive medium interactive-engagement level ($g = 0.50$, $d = 1.20$, $p < .001$).
382 PBL satisfaction was high and consistent ($M = 3.74$, $SD = 0.30$). Exhibition performance was strong
383 ($M = 82.62/100$, $SD = 9.98$, $N = 53$). Addressing RQ3: 84.9% of students adopted AI tools for simulation
384 coding, while mathematical derivation remained predominantly human-led, establishing an emergent
385 complementarity model that may serve as a template for AI integration policy in STEM PBL contexts.
386 The dominant breakthrough theme—linking analytical EOM derivation to simulation output (37.7%)—
387 triangulates with the largest-gain SE items, providing cross-strand validation of the framework's core
388 mechanism.

389 The SF-PBL framework provides a replicable, evidence-based, and OBE-compliant pedagogical model
390 for STEM educators seeking to authentically integrate computational simulation into engineering dynamics
391 teaching. The full dataset, survey instruments, rubric, and Python analysis scripts are available from the
392 corresponding author upon reasonable request.

CONFLICT OF INTEREST STATEMENT

393 The authors declare that the research was conducted in the absence of any commercial or financial
394 relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

395 SN: Conceptualisation, course design, SF-PBL framework development, data collection, formal analysis,
396 methodology, software, writing (original draft, review & editing), project administration. DD: Conce-
397 ptualisation, qualitative thematic synthesis, evaluation rubric design for behavioral and communication
398 dimensions, writing (review & editing). SS: Supervision, institutional resource allocation, curriculum
399 integration, methodology validation, writing (review & editing). KA: Academic oversight, OBE/NBA
400 alignment framework, validation of pedagogical outcomes, writing (review & editing).

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DATA AVAILABILITY STATEMENT

The anonymised individual-level survey dataset ($N = 53$, 65 variables), exhibition rubric evaluation scores (six judges across 17 teams), survey instruments, rubric specifications, Python analysis scripts (SciPy 1.11, NumPy 1.25), and representative student project artefacts are publicly available in the official repository: <https://github.com/sunny-nanade/SF-PBL-Engineering-Dynamics-2026>.

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APPENDIX A: SURVEY INSTRUMENTS

461 A.1 Self-Efficacy Scale (SE1–SE10)

462 *Instruction: Rate your confidence in each ability on a 5-point scale (1 = Strongly Disagree, 5 = Strongly*
463 *Agree).*

- 464 1.I can apply kinematic analysis to identify degrees of freedom in a multi-body system. [CO1]
- 465 2.I can construct a correct Free Body Diagram for a complex dynamic mechanism. [CO2]
- 466 3.I can derive the equations of motion for a system using Newton's Second Law. [CO3]
- 467 4.I can formulate the Lagrangian ($L = T - V$) for a conservative dynamic system. [CO4]
- 468 5.I can convert a second-order ODE system to state-space form ($\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$). [CO5]
- 469 6.I can implement a numerical ODE solver (`solve_ivp`) in Python to simulate a dynamic system. [CO5]
- 470 7.I can validate my simulation output using energy conservation principles. [CO5]
- 471 8.I can tune PID controller gains to achieve a specified dynamic response. [CO5]
- 472 9.I can correctly interpret simulation time-domain plots of position, velocity, and energy. [CO5]
- 473 10.I can communicate my simulation results clearly and accurately to a non-specialist audience.
- 474 [Communication]

475 A.2 Knowledge MCQ Items (PK1–PK5)

476 *Instruction: Select the single best answer.*

- 477 **PK1.** A rigid body constrained to rotate about a fixed axis in a plane has: (a) 3 DOF; (b) 1 DOF;
478 (c) 2 DOF; (d) 0 DOF.

- 479 **PK2.** The Lagrangian L is defined as: (a) $L = V - T$; **(b)** $L = T - V$; (c) $L = T + V$; (d) $L = \partial T / \partial q$.
- 480 **PK3.** The natural frequency of a simple pendulum of length l is: **(a)** $\omega_n = \sqrt{g/l}$; (b) $\omega_n = \sqrt{l/g}$;
481 (c) $\omega_n = g/l$; (d) $\omega_n = l/g$.
- 482 **PK4.** Adding viscous damping to an underdamped oscillator: (a) increases natural frequency; (b) has no
483 effect on frequency; **(c) decreases the oscillation frequency (damped natural frequency)**; (d) doubles
484 the amplitude.
- 485 **PK5.** Newton's Second Law for rotation states: (a) $\tau = ma$; **(b)** $\tau = I\alpha$; (c) $\tau = I\omega$; (d) $\tau = m\alpha$.